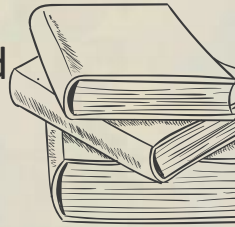
Personal Knowledge Graph Memory Engine and Career Development Assistant

Turning conversations into structured long-term memory for personalized and evolving career guidance



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01
Introduction

Motivation

The "Memory Gap" in Conversational AI

- Loss of depth and consistency across long-term interactions.
- Critical career signals are flushed out of the context window.

Limitations of Pure Vector RAG

- Flat similarity ignores complex relational context.
- Fails to capture the "Why" behind professional growth.

A "Brain + Expert" Architecture

Relational Memory (The "Brain")

Building persistent and evolving user personas through structured Knowledge Graphs.

Semantic Alignment

Seamlessly mapping unstructured conversational signals to industry-standard O*NET datasets.

Actionable Guidance (The "Expert")

Transforming memory into personalized gap analysis and professional development roadmaps.

02
Memory Architecture

Raw Conversation Data
ShareGPT-style multi-turn chats

Cleaning & Splitting
Turn-levei records, filter noisy data

Memory Extraction
Detect signals & extract memories

Memory Extraction

LLM Extraction (Groq/Llama-3.1):

Parsing unstructured dialogue into 7 structured signal types (Tools, Goals, Preferences, etc.).

Rule-Based Fallback:

Ensuring system reliability through deterministic keyword and pattern matching.

Knowledge Graph Storage

Relational Storage:

Utilizing NetworkX (MultiDiGraph) to preserve hop connections between users, projects, and goals.

Deduplication & Linking:

Using fuzzy string matching (SequenceMatcher) to merge conversational variants into canonical nodes.

Memory Dynamics & Retrieval

Dual-Weighting Retrieval:

- Mention Frequency: Calculating node importance based on frequent mentions and project connectivity.
- Temporal Recency: Automatic decay of outdated nodes used for time-sensitive memory ranking.

Contextual Subgraph Surfacing:

- Retrieving semantically relevant clusters rather than isolated keywords to maintain conversational context.

Robustness of the Memory Engine

Structural Fidelity ((Validated via Gold Standard)):

- **0.97 Relation F1**
  Demonstrating exceptional accuracy in preserving complex relational links and multi-turn conversational context.

- **0.88 Entities F1**
  Enabling high-precision extraction of career-critical nodes across diverse professional domains.

Internal Stability:

- **1.0 Noise Robustness**
  Maintaining perfect resilience and anti-interference against typos, conversational fillers, and colloquial speech patterns.

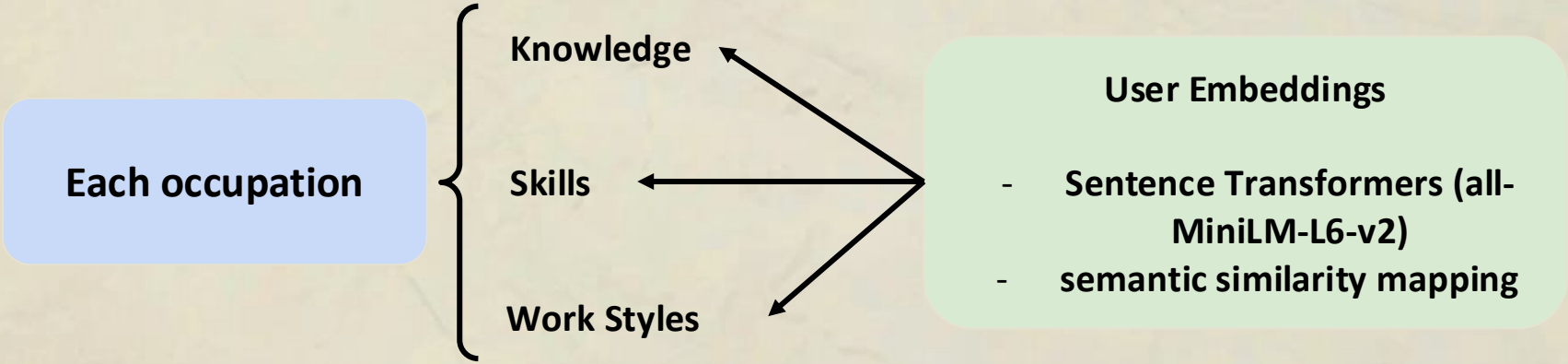
- **1.0 Self-Consistency**
  Ensuring deterministic and stable graph construction results across redundant and independent processing runs.

03
Career Recommendation Architecture

Overview

Conversations between user and AI → Signal extraction (skills / interests / projects /…) → O*NET profile alignment → Two-stage recommendation → Gap analysis

Industry Alignment & Semantic Mapping



Cognitive Inference Engine - Inferring Implicit Traits

LLM extracts explicit signals & System infers implicit abilities

Examples:

mentioned many technical projects	→	Complex Problem Solving
repeated debugging	→	Attention to Detail
mentioned heavy workload	→	Stress Tolerance

Two-Stage Career Recommendation

Stage 1 — Vector Retrieval

- weighted cosine similarity
- Top-20 candidates

Stage 2 — LLM Re-ranking

- Groq + full context
- Top-5 + reasoning

Actionable Feedback: Gap Analysis

Compare User profile vs Target occupation

Output:

- **missing skills**
- **missing knowledge**
- **development directions**

Example:

- **CS → + math / stats / finance → Quant**

04
Evaluation for Career Recommendation Engine

Evaluation Framework & Results

AI-as-a-Judge

We leverage GPT-4o as an 'authoritative judge' to provide automated, expert-level evaluation for our Groq-extracted career data, overcoming the lack of ground-truth labels through high-reasoning semantic analysis.

- **Qualitative scoring: Completeness, Faithfulness, and Career Utility**
- **Narrative-based profile validation: Generating a "Career Profile Narrative" to ensure the Agent's understanding matches user intent**

Results

GPT-4o Career Judge
- Completeness: 3.133 / 5
- Faithfulness: 3.3 / 5
- Career utility: 3.133 / 5

Results

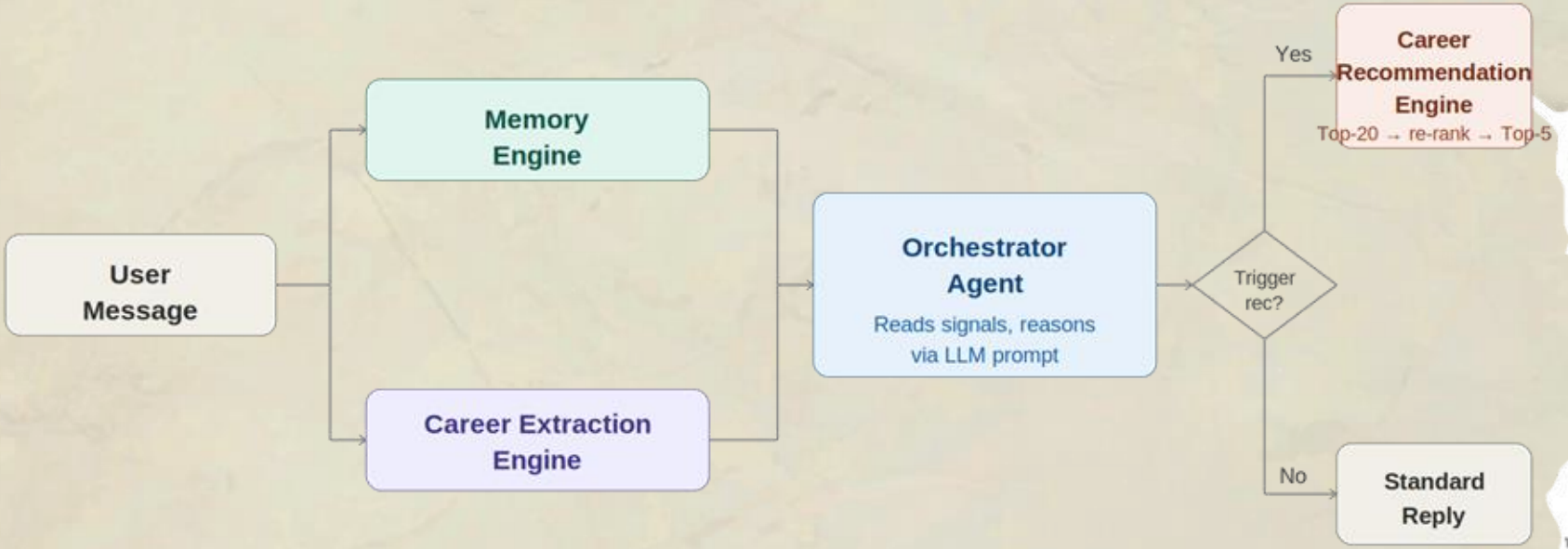
Narrative-based profile validation examples:

"turn_id": "career_dev_198",
"completeness": 3,
"faithfulness": 2,
"career_utility": 3,
"rationale": "Some key signals like **interest in portfolio optimization** and f**ast-paced work style** are missing or misrepresented, and some extracted items are **not grounded in the conversation**."

"turn_id": "career_dev_145",
"completeness": 4,
"faithfulness": 5,
"career_utility": 4,
"rationale": "The profile **captures most key signals from the conversation**, is well-grounded in the text, and provides **actionable insights**, though it could include more specific career path suggestions."

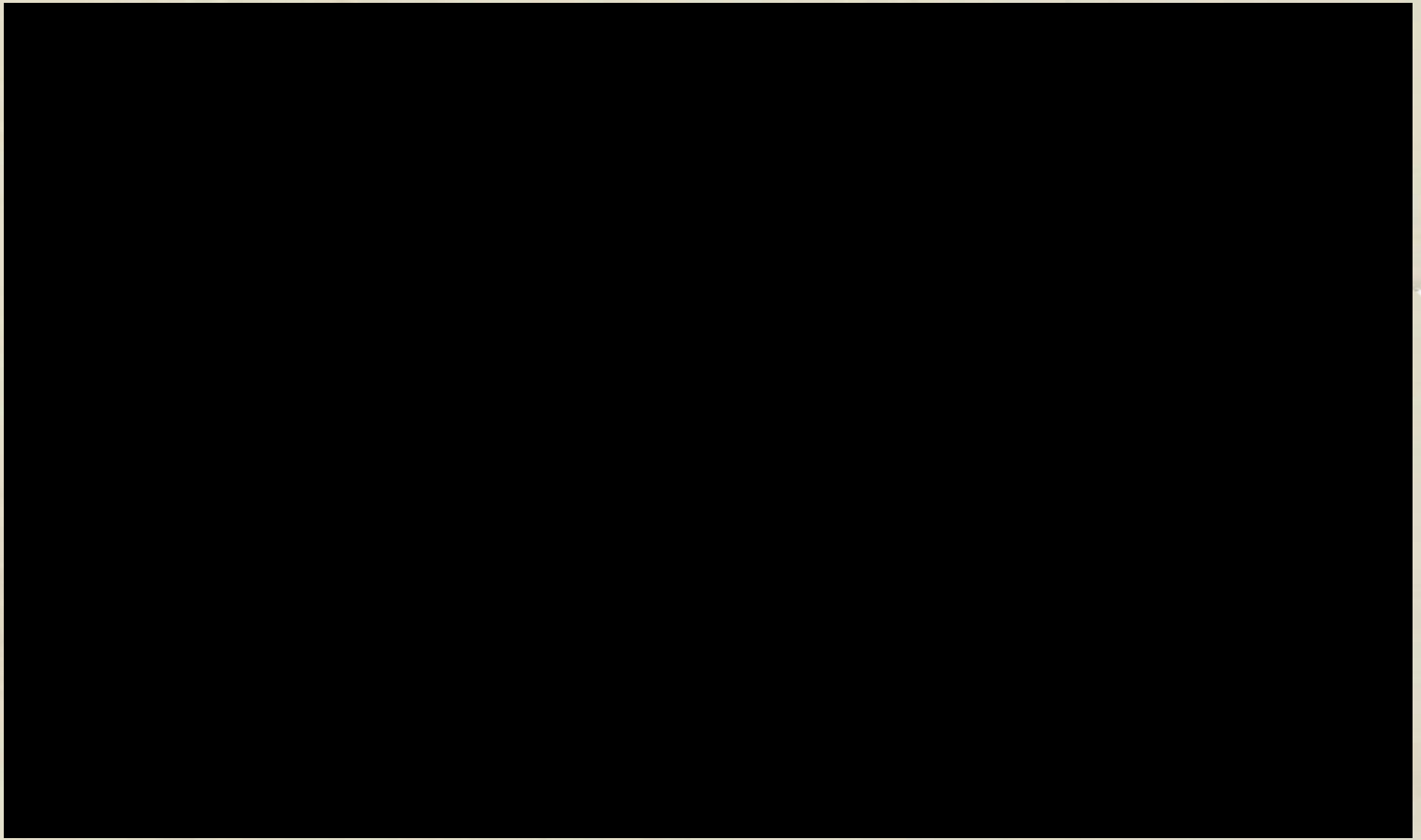
05
Agentic Orchestration
& Live Demo

Agentic Orchestration — Connecting the Brain and the Expert



Demo

Demo



06
Conclusion

Key Takeaways

The Power of Structured Memory

- Bridging Objective Data (O*NET) and Subjective Reasoning (LLM).
- Creating a coherent, evolving persona from unstructured dialogue.

Key Project Impact

- Beyond simple Q&A: Persistent decision support for career growth.
- High Precision: 0.97 Relation F1 score and 4.31 Faithfulness. (Memory Engine)

Future Vision

- Dynamic Embedding alignment and Interactive Graph Visualization.
- Human-in-the-loop: User-centric memory refinement.

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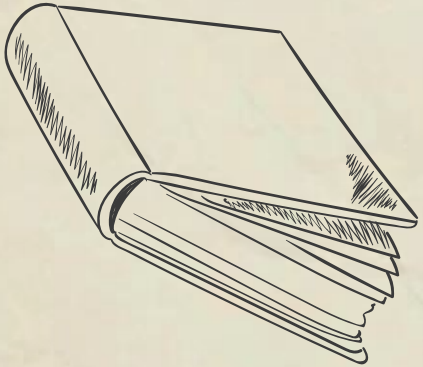

Thanks!